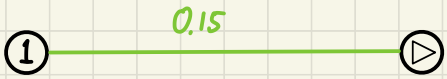
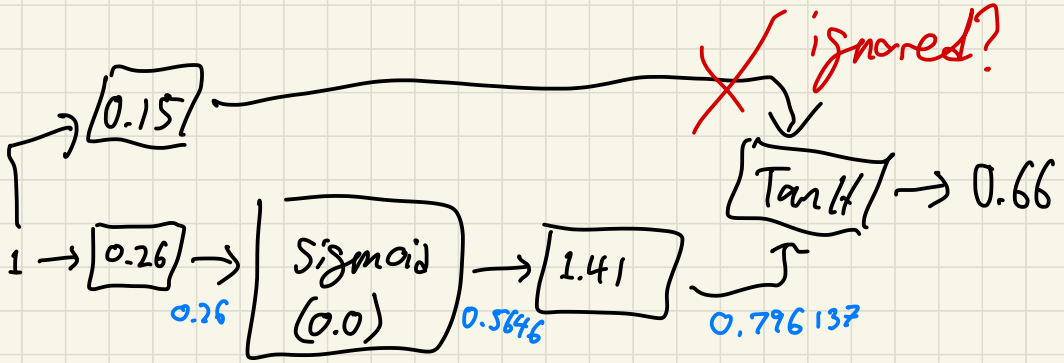
Iratus Populator





$$\sigma(x) \approx \frac{1}{1+e^{-x}}$$

$$\tanh(x) = \frac{e^{2x}-1}{e^{2x}+1}$$

$$\operatorname{arctanh}(0.66) \approx 0.792814$$

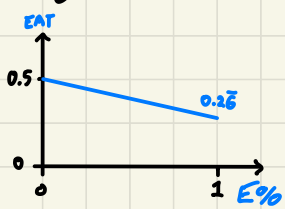
$$A \approx \tanh(0.\bar{6})$$

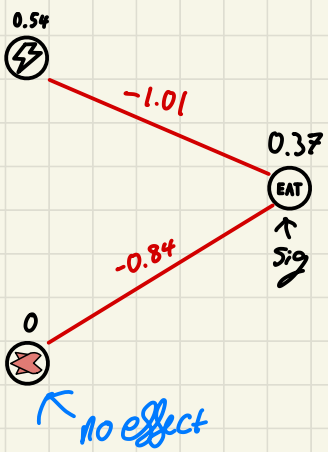
empty world output node values:



00:00:19 → 0.33 (1)
00:00:32 → 0.37 (1)
$\frac{0.04}{13\ s} \approx 3.08\ m/s$

$EAT \approx sig(-1.01(E\%) - 0.84)$





0.267 min

$f(1) = 0.2\overline{6}$

$f(0) = 0.5$

$f(x) = mx + b$

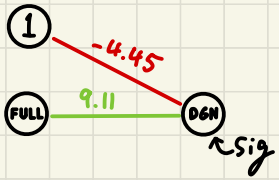
$b = \frac{1}{2}$

$\frac{4}{15} = m + \frac{1}{2}$

$\frac{8}{30} - \frac{15}{30} = \frac{-7}{30} = m$

$EAT \approx \frac{-7}{30}(E\%) + \frac{1}{2} \rightarrow EAT \approx sig(-1.(E\%) - 0.85)$

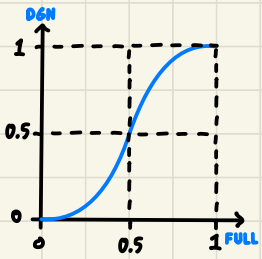
→ digestion:



$$DGN = sig(9.11(FULL) - 4.45)$$

$$DGN(0) \approx 0.011544$$

$$DGN(1) \approx 0.990622$$



↳ $DGN \approx sig(10(FULL) - 5)$

$$\frac{d}{d(full)} DGN(full) = \frac{9.11 e^{4.45 - 9.11x}}{(e^{4.45 - 9.11x} + 1)^2}$$

$$\frac{d}{d(full)} DGN(full) = 2.27123 \text{ at } full = \frac{1}{2}$$

Plants & rotation

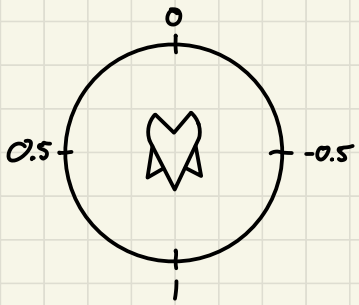
1.88 → tanh ROT ≈ tanh(1.88(plant Δ))

Red Pheremone angle → 2.78 → Σ → tanh → Herd

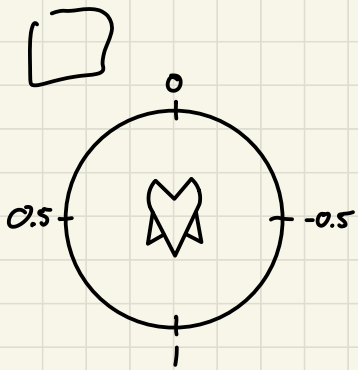
-0.5 → Σ

Plant (closeness) → -1.37 → Gauss → 1.42 → tanh → Grab

2.78 − 0.5 (



[(-1.37)(0.83)] → gauss → 0.44



↳ assuming angle out is directly proportional to value in required to make bibite turn towards the stimulus.

↳ rotation is clockwise?

↳ yes, confirmed experimentally.

0.25 → rot

OH. so strangely enough, the red pheremone interaction is directional.

WAIT: it actually doesn't even matter. functionally zero impact?

→ minute is being used as timeline;

$Relu_{in} = (-0.44)timealive + (1.61)NPlants + (-0.85)Fullness$

$Relu_{in} = 1.61 N_{plants} - 0.85\,fullness - 0.44\,timealive$

$1.61 = 0.44\,timealive$

$timealive = 3.66\ sec$

↳ Iratus Populator emits red pheromones while younger than 3.66 seconds per nearby plant while hungry.

↳ when full, this limit decreases to $1.\overline{72}$ seconds per nearby plant.

Grab



$$GRB = \tanh\left(1.42\left(\text{gauss}\left(-1.37\left(\text{PlantCloseness}\right)\right)\right)\right)$$

$$GRB \approx \tanh\left(-1\cdot\left(\text{PlntΔ}\right)+\frac{3}{2}\right)$$

0.00 → 0.91
0.02 → 0.90
0.06 → 0.89
0.13 → 0.88
0.17 → 0.87
0.20 → 0.86
0.24 → 0.85
0.28 → 0.84
0.32 → 0.83
0.35 → 0.82
0.39 → 0.80
0.43 → 0.79
0.46 → 0.78
0.54 → 0.75
0.57 → 0.73
0.61 → 0.71
0.65 → 0.69
0.68 → 0.67
0.72 → 0.65
0.76 → 0.63
0.79 → 0.61
0.83 → 0.59
0.87 → 0.56
0.90 → 0.54

data matches model! ✓

Artificial Iratus Populator

↳ $A \approx 0.6\bar{} \approx \operatorname{arctanh}(0.8)$

↳ $EAT \approx \text{Sig}(-1 \cdot (E\%) - 0.85)$

↳ $DGN \approx \text{Sig}(10(\text{FULL}) - 5)$

↳ $ROT \approx \tanh(1.88(\text{PlntΔ}))$

↳ $HRD \approx \tanh(0.\bar{3}(\text{PlntΔ}) - \frac{1}{2})$

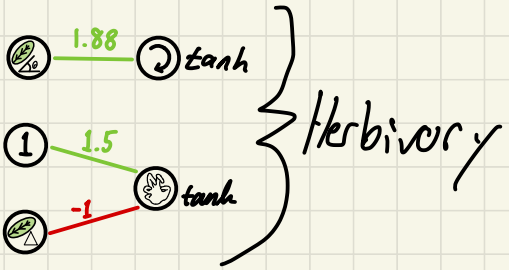
↳ $ATK \approx 2.25(\text{Bibite Closeness})$

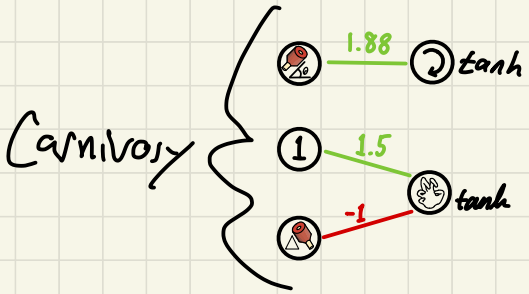
↳ $GRB \approx \tanh(-1 \cdot (\text{PlntΔ}) + \frac{3}{2})$

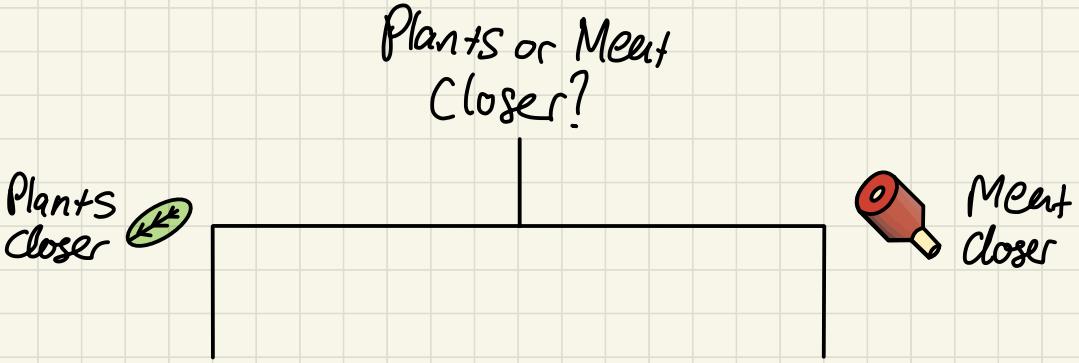
"Iratus Artificium v1"

v2

Omnivory







Plants closer = -1

Meat closer = +1

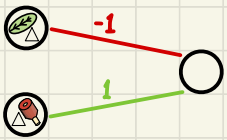
equidistant = 0

Plants closer = -1

Meat closer = +1

equidistant = 0

isEating?

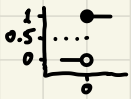


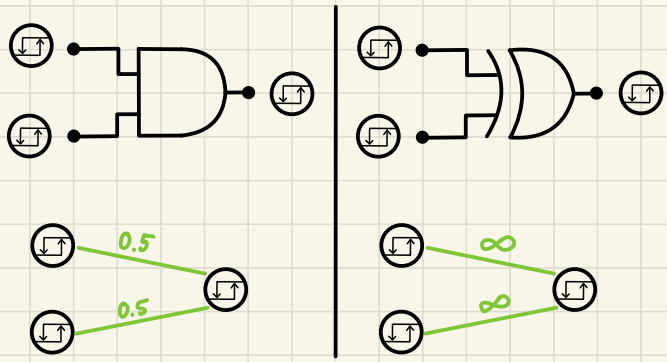
$X = 0.5 + Rel(\infty \times (Diff($










0.5
0.5
∞
∞

diff $0^- = 100$

diff $0^+ = 0$

$0 \to 0.01^- \to 100$

$0.01^- \to 0.01^+ \to -0.04$

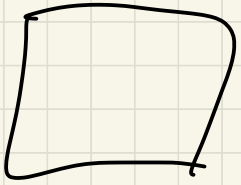


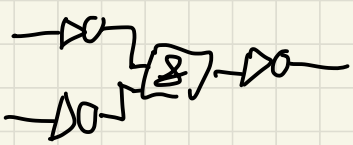
A	B	Σ	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

1	0	0	0
0	1	1	1
1	0	0	0

0	0	1
0	1	0
1	0	0
1	1	1







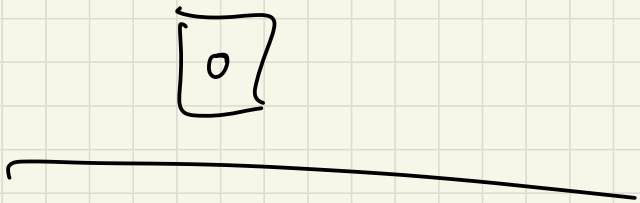
5

8	0
4	1
2	0
1	1

→

7

8	0
4	1
2	1
1	1



Archival note 10/21/2026: this note was dated in my Goodnotes app as being from 08/09/2023 -Steve